Global Invest Bank: A Web-Based Banking System with MetaMask Integration and Interactive Security Features

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***Abstract*—This project presents the design and implementation of a web-based banking system called Global Invest Bank. The system integrates traditional banking features such as account creation, login authentication, and balance management with modern blockchain technology using MetaMask. Additionally, interactive security mechanisms, including puzzle-based authentication, are implemented to enhance user engagement and prevent unauthorized access. The system is developed using HTML, CSS, JavaScript, Node.js, and MySQL, providing a full-stack solution for secure and user-friendly financial interaction.**

***Index Terms*—Web Development, Blockchain, MetaMask, MySQL, Node.js, Authentication, UI Design**

I. Introduction

The evolution of web-based financial systems has led to increased demand for secure, interactive, and user-friendly platforms. Traditional banking applications focus heavily on security but often lack engaging user interfaces. This project aims to bridge that gap by combining a visually appealing design with functional financial operations and blockchain integration.

Global Invest Bank is a web-based application that allows users to create accounts, manage balances, and interact with MetaMask wallets. The system also introduces puzzle-based authentication mechanisms to enhance security in a creative and interactive way.

II. System Design

The system follows a client-server architecture consisting of a front-end user interface and a back-end server connected to a MySQL database.

A. Front-End

The front-end is built using:

- HTML for structure
- CSS for styling and animations
- JavaScript for interactivity

Key features include:

- Girly aesthetic UI design with gradients and animations
- Interactive puzzle-based login and signup
- Real-time balance updates
- MetaMask connection interface

B. Back-End

The back-end is implemented using Node.js and Express. It handles:

- User authentication
- Session management
- Communication with MySQL database

C. Database

The MySQL database contains:

- User credentials (userid, password, etc.)
- Financial data (savings balance)

III. MetaMask Integration

MetaMask is used to connect blockchain wallet functionality to the system. The integration allows users to:

- Connect their wallet
- Retrieve wallet address
- Retrieve Sepolia ETH balance
- Convert blockchain balance into application currency

The system uses JavaScript API calls such as:

```
window.ethereum.request({
  method: 'eth_requestAccounts'
});
```

and

```
window.ethereum.request({
  method: 'eth_getBalance'
});
```

IV. Security Features

To improve security and user interaction, puzzle-based authentication is implemented.

A. Puzzle Login

Users must solve a drag-and-drop puzzle before accessing the login form.

B. Puzzle Signup

Similarly, account creation is locked until users complete a matching puzzle game.

This reduces automated attacks and adds an engaging layer of security.

V. Functionality

The system supports the following features:

- Account creation
- Login authentication
- Deposit and withdrawal operations
- Real-time balance updates
- MetaMask wallet connection
- Blockchain balance integration

VI. Results

The system successfully demonstrates:

- Full-stack web application development
- Integration of blockchain technology into a traditional system
- Enhanced user experience through UI design and gamification

Testing showed that:

- Users can securely log in and manage balances
- MetaMask integration retrieves wallet data correctly
- Puzzle authentication prevents immediate access

VII. Conclusion

Global Invest Bank demonstrates how modern web technologies and blockchain integration can be combined to create an interactive and secure financial application. The addition of puzzle-based authentication enhances both usability and security. Future improvements may include real transaction handling, smart contract integration, and advanced encryption methods.

VIII. Future Work

Future enhancements may include:

- Smart contract-based transactions
- Real cryptocurrency transfers
- Multi-user real-time synchronization
- Enhanced security using encryption techniques

References

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